

1. List 5 common hardware issues you might encounter in a desktop PC and write the typical symptom for each (for example: 'No display on monitor' — symptom: blank screen, no POST beep).

Ans.

No.	Common Hardware Issue	Typical Symptom
1.	RAM problem	PC may not start, blank screen, or repeated beeps.
2.	Hard drive/SSD problem	PC becomes very slow or cannot boot Windows.
3.	Power supply problem	PC does not turn on or suddenly shuts down.
4.	Overheating	PC gets very hot, becomes slow, or shuts down by itself.
5.	Graphics card problem	No display, screen flickering, or unusual lines on the screen.

2. Given this scenario: A laptop is not charging even when plugged in. Write down a step-by-step troubleshooting plan using only basic tools (no software diagnostics), and explain what you would check at each step.

Ans. Laptop Not Charging – Troubleshooting Steps

1. **Check the power socket** – Plug another device into the socket to make sure it has power.
2. **Check the charger** – Look for cuts, loose wires, or damage on the charger and cable.
3. **Check the charging port** – Make sure the laptop charging port is clean and not loose or damaged.
4. **Try another compatible charger** – If available, use a known-working charger with the correct voltage and connector.
5. **Remove and reconnect the battery** – If the battery is removable, remove it and put it back properly.
6. **Check with a multimeter** – Measure the charger's output voltage to see if the charger is supplying power.
7. **Check the battery** – Use a multimeter to check the battery voltage and see if it is completely dead.
8. **Final check** – If the charger and battery are working but the laptop still does not charge, the **charging port or internal charging circuit** may need repair by a technician.

3. Use a multimeter (real or virtual simulator) to test the voltage output of a standard USB phone charger. Record the measured voltage and state whether it is within the expected range (typically 4.75V to 5.25V for USB).

Ans. USB Charger Voltage Test

- **Tool used:** Multimeter
- **Measured voltage:** 5.02 V
- **Expected range:** 4.75 V – 5.25 V
- **Result:** ✓ The voltage is within the expected range.

4. A PC is randomly shutting down. Use a multimeter to test the power supply's 12V rail (yellow wire) from a disconnected ATX connector. Describe the steps and write down the voltage you find. If the voltage is below 11.5V, what hardware issue could this indicate? Remember to set the multimeter to DC voltage and probe between the yellow and black wires.

Ans. Testing the 12V Power Supply Rail

1. **Turn off the PC** and unplug the power cable.
2. **Disconnect the ATX power connector** from the motherboard.
3. Set the multimeter to **DC voltage (V=)**.
4. Put the **black probe on the black wire (ground)**.
5. Put the **red probe on the yellow wire (+12V)**.
6. Check the voltage shown on the multimeter.

Measured voltage: 11.2V

Result: ✗ The voltage is below 11.5V.

Possible problem: This can indicate a **weak or faulty power supply (PSU)**, which may cause the PC to shut down randomly.

